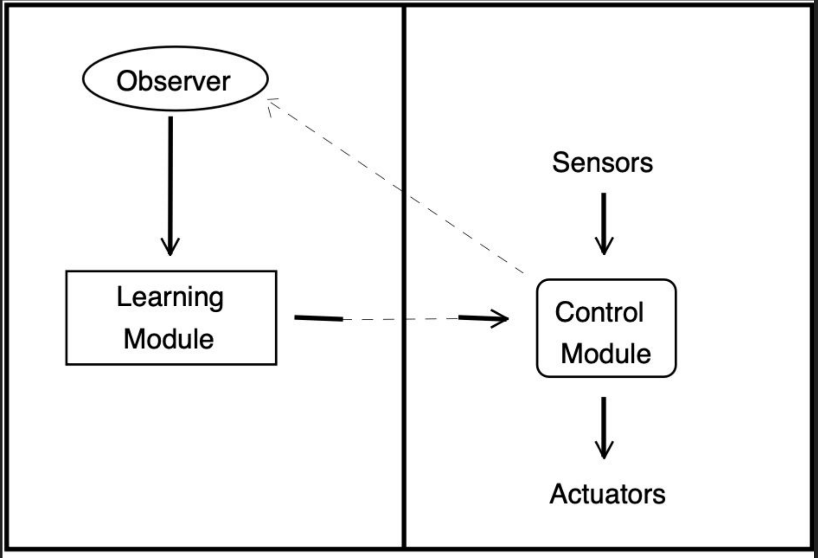
Evolving Area Coverage Using a CGA and Punctuated Anytime Learning on a Real Robot

By Faiz Aladin

Advisors: Gary Parker, Jim O’Connor

What is Punctuated Anytime Learning?

Training a robot on an offline simulation and periodically updating the simulation and improving the learning module by testing the learning module on an actual robot.



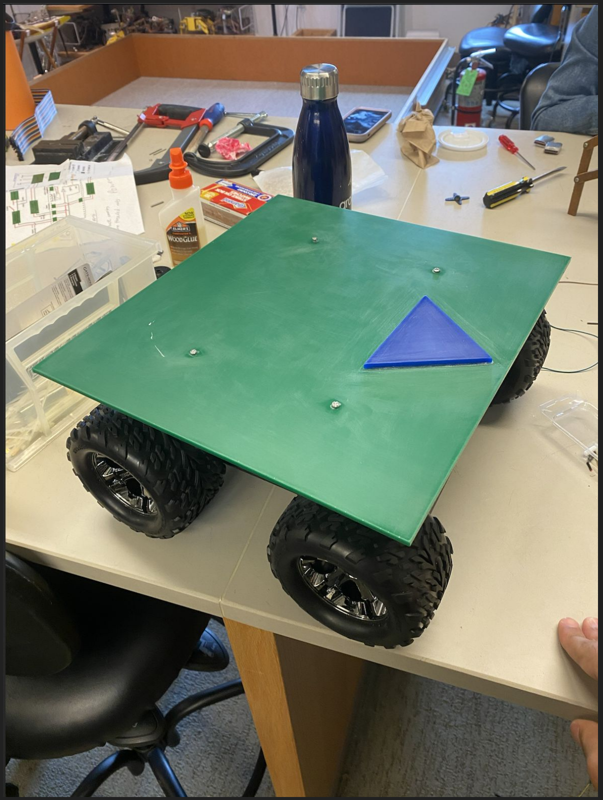
Problem Description

How do we teach a 4 wheeled robot to effectively cover an area?

The Experiment:

- Have the robot start on one end of the colony and traverse the area in snake pattern
- Assuming each grid square has a number, it must be in sequential order

The Robot



How does the Robot see?

```python
def reset():

    buffer_size = 20

    # Initialize the camera
    cap = cv2.VideoCapture('rtsp://admin:123456@136.244.195.47:554/Streaming/channels/0')  # Use 0 for the default camera

    cap.set(cv2.CAP_PROP_BUFFERSIZE, buffer_size)

    try:
        ret, frame = cap.read()
        robot_info = testcamera.calculate_orientation(ret, frame)

        while robot_info[1] > 5 and robot_info[1] < 355:
            #robot_info = testcamera.calculate_orientation(ret, frame)
            print(robot_info[1])
            if robot_info[1] < 5 or robot_info[1] > 355:
                break
            elif robot_info[1] > 90:
                move.right(random.uniform(0.15, 0.4), 0.85)
                move.stop()
            else:
                move.left(random.uniform(0.15, 0.4), 0.85)
                move.stop()
            time.sleep(2)
            print("checking")
            cap = cv2.VideoCapture('rtsp://admin:123456@136.244.195.47:554/Streaming/channels/0')  # Use 0 for the default camera
            cap.set(cv2.CAP_PROP_BUFFERSIZE, buffer_size)
            ret, frame = cap.read()
            robot_info = testcamera.calculate_orientation(ret, frame)
```

Prerequisites

Robot’s center and orientation

- Tells us the location of the robot and which way it is facing (0-360)

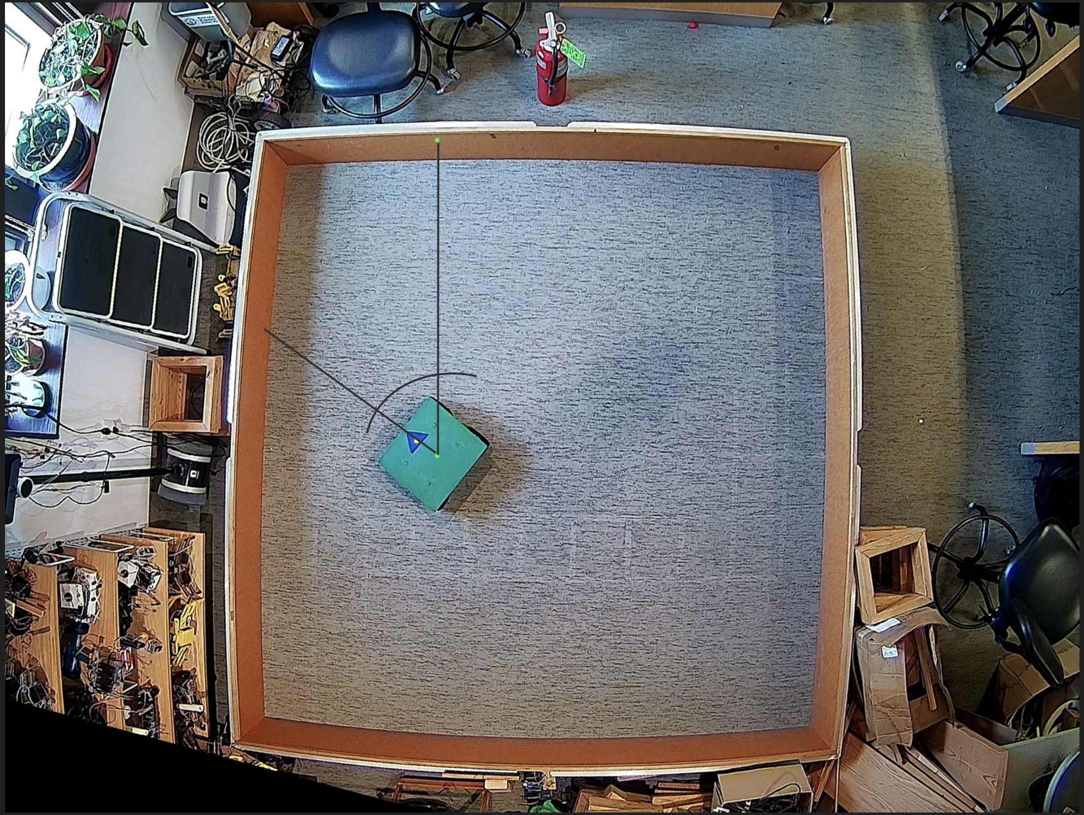
Grid Tracking

- Tells us which grid squares the robot has traveled in

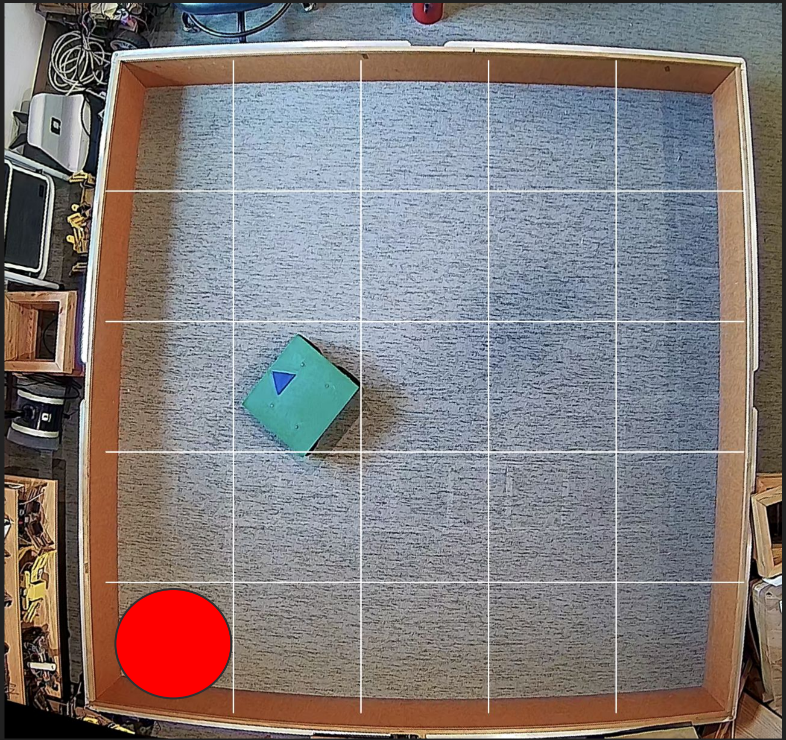
Reset

- Robot should be able to reset to starting position after testing a chromosome

Robot’s Center and Orientation



Grid Tracking



Reset

Method

Use a Cyclic Genetic Algorithm to teach the robot the how to cover the area.

- Have the robot cover the area in simulation and test on an actual robot every punctuated generation.

Cyclic Genetic Algorithm

Similar to a GA except each chromosome represents a cycle of tasks

- Each gene (group of bits) represents an action and the number of times it is repeated

Phase 1:	1	000	1	00	
Phase 2:	0	110	0	10	010
	Direction of movement	Amount of Curve	Direction of Turn	Amount of Turn	Number of loops
	0 - right 1 - left	8 levels of curve	0 - right 1 - left	4 degrees of turn	0-7 possible loops

Fitness Function

We calculate the fitness by taking the sum of sequential grid squares covered by the robot starting from 1.

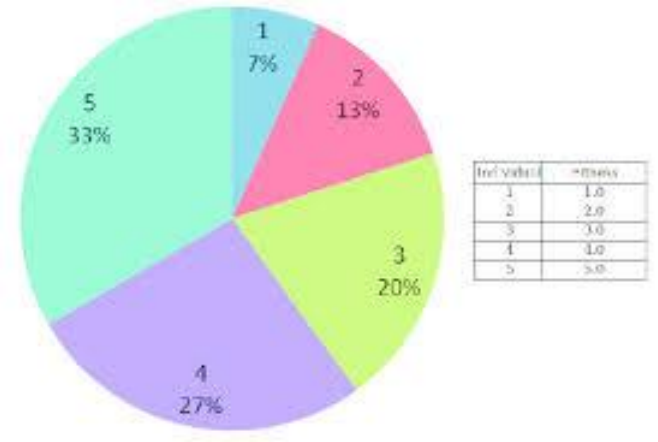
Specifics of the CGA

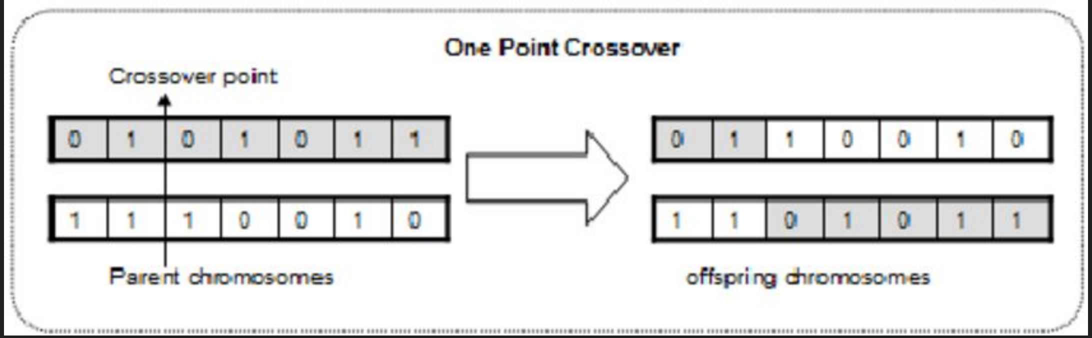
Mutation Rate: 0.01

Population Size: 100

Generations: 5000

Selection Style: Roulette Selection





Next Steps

Implement a simulation for the task

- Test the most fit individual on the real robot every 500 generations
- Send feedback to the learning module

References

1. Parker, Gary. (2002). Punctuated anytime learning for hexapod gait generation. IEEE International Conference on Intelligent Robots and Systems. 3. 2664 - 2671 vol.3. 10.1109/IRDS.2002.1041672.

2. Blumenthal, H. & Parker, Gary. (2004). Punctuated Anytime Learning for Evolving Multi-Agent Capture Strategies. 2. 1820 - 1827 Vol.2. 10.1109/CEC.2004.1331117.

3. Parker, Gary & Fedynyshyn, Gregory. (2007). Enhancing embodied evolution with punctuated anytime learning. 190-195. 10.1109/ICSMC.2007.4413720.

4-10. Talked to Gary Parker